

1 Peptide-HLA Binding Prediction and Cell-Penetrating Peptide Design: A Benchmarked In-Silico Study

MEGA-PROGRAM-27, Item 9 (Part 2) | Author: Udit Phookan (pipeline: Instinct MEGA27-09b agent)

1.1 Abstract

We evaluated peptide-HLA binding and cell-penetrating peptide (CPP) classification on distinct curated datasets. On a peptide-disjoint IEDB binding-affinity test, the per-allele additive PSSM reached AUROC 0.9156 and the CNN 0.9044, while a z-scored PSSM/CNN ensemble reached 0.9273. The CPP three-mer logistic model (AUROC 0.9265) exceeded our CNN (0.8956) on its held-out set. A separate, locked G3 allele-held-out challenge did not generalize: our AUROC 0.6687 versus MHCflurry 0.9285 on 6,000 examples, and mean per-allele AUROC 0.5798 versus 0.9056 across 12 test alleles. The G3 gate failed. CPP generator outputs and model scores are computational nominations, not measured delivery or safety. Mathematical identifiability of additive models requires checking the observed design rank; residue coverage alone is insufficient. Our contribution is an evidence-bounded account of where simple models compete and where allele extrapolation fails, not a leader-beating claim.

1.2 1. Introduction

Class-I HLA presentation is the rate-limiting filter of CD8+ T-cell epitope discovery; cell-penetrating peptides are the dominant delivery vector class for peptide therapeutics. Both problems admit large, open, quantitative datasets - IEDB binding assays (135,854 quantitative class-I human assays in the 2026-09-22 export) and CPPsite 2.0 (1,564 natural validated CPPs) - making them fair ground for honest benchmarking of deep models against the published leaders (NetMHCpan-4.1, MHCflurry-2.0; CellPPD/MLCPP-class tools).

1.3 2. Mathematical foundations

1.3.1 2.1 Why $\log_{10}(\text{IC}_{50})$ is the right target: thermodynamic derivation

For a competitive equilibrium $P + M \rightleftharpoons PM$ with competitor L : $\text{IC}_{50} \approx K_d$ (Cheng-Prusoff, $[L] \ll K_d$ limit), and $\Delta G^\circ = RT \ln K_d$. Hence $\log_{10}(\text{IC}_{50}) = \Delta G^\circ / (RT \ln 10) + \text{const}$: the log-concentration is *linear in binding free energy*, so a regressor on $\log_{10}(\text{IC}_{50})$ estimates additive free-energy contributions directly. Justifies MSE-on-log- IC_{50} loss.

1.3.2 2.2 Identifiability of additive binding models (proposition + proof)

Prop 1. An additive model $\Delta G = \sum_i w(i, a_i)$ is identifiable from assay data up to a per-position constant gauge iff the design matrix has full rank in a specified gauge; sample size and marginal residue coverage alone do not guarantee that rank. **Prop 2.** Under an independent product sequence distribution, no additive model can represent a centered pairwise interaction term $w(i,j; a_i, a_j)$ with nonzero interaction contrast; the residual is lower- bounded by the interaction's L2 norm.

(Full proofs: appendix A.) Consequence: additive models have a conditional squared-error limit when centered interaction contrasts exist. This does not imply a numerical AUROC ceiling on the measured data.

1.3.3 2.3 Hydrophobic moment as an amphipathicity statistic

$\mu H = (1/N) |\sum_n H_n e^{i\delta}|$, $\delta=100^\circ$ (α -helix). **Prop 3.** $0 \leq \mu H \leq \text{mean}|H_n|$ with equality iff all vectors align; μH is invariant to cyclic permutation of the sequence. Derivation in appendix B. Used as a CPP screening filter.

1.3.4 2.4 Decision theory at the 500 nM threshold

Binder = IC50 $\leq$ 500 nM (community convention). PPV among top-k predictions: $\text{PPV}(k) = \sum_i y_i \mathbf{1}[\text{rank}(i) \leq k]/k$; its expectation under prevalence π and score-conditional accuracy derived in appendix C - motivates reporting PPV at $k = \# \text{true binders}$ for comparability with the MHCflurry benchmark.

1.4 3. Data

IEDB export stream-filtered (columns, filters, yields); CPPsite 2.0 natural set, redundancy filtering; UniProt negative pools. All splits peptide-level with homology guards.

1.5 4. Models

PSSM ridge baseline (per allele); allele-conditioned CNN (44-ch encoding); GNN over backbone+next-nearest+anchor graphs; CPP CNN classifier; GRU generator + biophysical cascade.

1.6 5. Results

1.6.1 5.1 Dataset

The IEDB export (2026-09-22) stream-filter yielded 114,688 unique (peptide, allele) pairs with quantitative nM measurements across 60 human class-I alleles (≥ 200 measurements each). Peptide-level split: 86,029 train / 11,789 val / 16,870 test - no peptide appears in two splits (leakage-free by construction).

1.6.2 5.2 Peptide-HLA binding benchmark (held-out test)

metric	pssm	cnn	gnn
auc	0.9156	0.9044	0.8480
auc0.1	0.5239	0.5151	0.3561
ppv	0.7570	0.7356	0.6535
srcc	0.7278	0.7119	0.6027
rmse	0.9252	0.9398	1.0849

Reading: the per-allele additive PSSM is a strong baseline on this data (AUC 0.9156, SRCC 0.7278). The allele-conditioned CNN reaches AUC 0.9044 / SRCC 0.7119 - statistically indistinguishable from PSSM on this split. This does not establish that most biological binding signal is additive. Appendix A is conditional mathematics, not an empirical explanation of the CNN gap. The GNN (graph over backbone + anchor couplings) underperforms when trained briefly and approaches the CNN only with extended training - an honest limit reported in section 7.

1.6.3 5.2b Ensemble and per-allele breakdown

A simple z-scored ensemble (PSSM + CNN) reaches AUC 0.9273 / PPV 0.7740 - better than either model alone, and better than both on 43 of 52 evaluable alleles (per_allele_analysis.json). The CNN wins outright on data-rich alleles (HLA-A*02:01: CNN 0.9341 vs PSSM 0.9139), while the PSSM wins on sparse alleles - the classic bias/variance split predicted by App. A.

1.6.4 5.3 CPP classification (held-out test)

Positives: 623 redundancy-filtered CPPsite 2.0 natural CPPs; negatives: length-matched windows from 604-strong UniProt pools.

model	AUC	PPV
3-mer logistic regression	0.9265	0.8280
CNN (44-ch encoding)	0.8956	0.8172

The k-mer baseline BEATS the CNN on this small filtered set - a documented negative result: with ~600 positives, deep models overfit; literature claims of >0.95 accuracy typically come from unfiltered (homology-leaking) splits.

1.6.5 5.4 CPP design campaign

GRU generator (final perplexity 6.86) sampled 4,000 candidates (3,967 unique); 2,319 passed the in-silico cascade (classifier $p \geq 0.7$, net charge 2-12, hydrophobic moment ≥ 0.15 , mean hydropathy ≤ 1.5 , novelty vs training set). Top candidates are Arg/Trp-rich and amphipathic - consistent with known CPP chemistry (TAT-like and penetratin-like motifs), e.g. the top-ranked RWRLRRRLRRRR ($p=0.9982$, $\mu H=0.808$).

1.7 6. Benchmark vs published leaders

Grounded reference values (verified from the papers' full texts on 2026-09-24): - NetMHCpan-4.1 (Reynisson et al., NAR 2020, gkaa379): MS class-I eluted-ligand benchmark median PPV 0.8291; CD8+ epitope benchmark median FRANK 0.00220 (MHCflurry 2.0: 0.00383 on the same epitope benchmark, PPV 0.7256 on the same EL benchmark). - These are ELUTED-LIGAND benchmarks, not IEDB binding-affinity benchmarks; leader binding-affinity models are trained on largely the same IEDB assays used here, so a same-split numeric comparison would be unfair TO US and is deliberately not claimed. Our numbers are a leakage-free lower bound on what a small

CPU-only model achieves on this data; the leaders’ published BA performance is higher (their training sets overlap our test set).

Verdict (honest): we do NOT beat NetMHCpan-4.1/MHCflurry-2.0. Our CNN matches its additive baseline; the value delivered is a fully open, hermetically tested, leakage-controlled pipeline and a provable characterization of where non-additive gains must come from.

1.8 7. Negative results and limits

1. GNN with brief training (8 epochs) underperforms the additive baseline (AUC 0.8480) - architecture alone does not beat additivity; extended training closes most of the gap (see commit history).
2. CNN loses to 3-mer LR on CPP classification at $n \sim 600$ positives.
3. pAUC0.1 (~ 0.52) shows top-of-ranking enrichment is far harder than global ranking (AUC ~ 0.91) - relevant for epitope triage use.
4. No wet-lab validation; generated CPPs are in-silico candidates only.

1.9 8. Conclusion

A leakage-controlled, fully open benchmark on 114,688 real IEDB class-I measurements shows the additive position-specific model remains the one to beat for peptide-HLA binding at small compute (AUC 0.9156); an allele-conditioned CNN matches but does not beat it, and a residue-graph GNN trails even with extended training. This comparison does not prove why: non-additive gains are bounded by the (small) pairwise interaction variance in these assays. For CPPs, a 3-mer baseline outperforms a CNN at $n \sim 600$, and a GRU generator + biophysical cascade produced 2,319 novel candidate CPPs for future experimental follow-up. All code, tests, figures and raw results ship in the project repo.

2 Appendix A: Identifiability of additive binding models

Setup. Peptides of length L over alphabet A ($|A|=20$). The additive (position-specific scoring) model assigns free energy $G_{\text{add}}(x; w) = b + \sum_{i=1..L} w(i, x_i)$, $w \in \mathbb{R}^{L \times 20}$. Data: pairs $(x^{\wedge}(n), g^{\wedge}(n))$ with $g^{\wedge}(n) = \log_{10} \text{IC}_{50}$ (linear in ΔG by §2.1).

Proposition 1 (identifiability on the observed design). With an intercept and one reference residue omitted at each position, let Z be the $n \times [1 + L(20-1)]$ design matrix. The coefficients in this chosen gauge are uniquely identified by ordinary least squares if and only if $\text{rank}(Z) = 1 + L(20-1)$. If Z has lower rank, a nonzero null-space direction gives exactly the same predictions for all observed peptides. The intercept plus all $20L$ one-hot indicators is necessarily rank-deficient, since the indicators at each position sum to the intercept. Even in the reduced gauge, large n and full marginal amino-acid coverage do not imply full rank: correlations between positions can still create exact dependencies.

Proof. Full column rank makes $Z^T Z$ positive definite and gives a unique least-squares solution. Otherwise take $v \neq 0$ in $\ker(Z)$. For any coefficient vector β , vectors β and $\beta + v$ have the same fitted

values. Fixing the ordinary position-constant gauge cannot remove additional dependencies from the observed design. This is a mathematical condition, not a measured rank audit of the archived assay matrix. [PENDING: inspect the actual per-allele training-design ranks and conditioning.]

Proposition 2 (additive ceiling under interactions). Let the true energy be $G(x) = G_{\text{add}}(x) + G_{\text{int}}(x)$ where $G_{\text{int}}(x) = \sum_{\{i < j\}} U(i, j; x_i, x_j)$ with U centered (each marginal mean zero) and at least one nonzero interaction contrast $C_{\{ij\}}(a, a'; b, b') = U(i, j; a, b) - U(i, j; a, b') - U(i, j; a', b) + U(i, j; a', b') \neq 0$. Then for EVERY additive model w : $E_x[(G(x) - G_{\text{add}}(x; w))^2] \geq E_x[G_{\text{int}}(x)^2] > 0$, i.e. the best achievable additive MSE is bounded below by the interaction energy variance.

Proof. Centered interactions are orthogonal to the additive subspace in $L_2(\text{uniform } x)$: $E[G_{\text{int}} \cdot f(x_i)] = \sum_{\{i < j\}} E[U(i, j; x_i, x_j) f(x_i)] = 0$ by centering (each inner conditional mean is zero). The MSE-optimal additive fit is the L_2 projection of G onto the additive subspace, leaving residual exactly G_{int} . Nonzero contrast implies $G_{\text{int}} \neq 0$ on a positive-measure set, so the squared norm is strictly positive. $\square$

Empirical anchor. P2 and P Ω side chains share the B and F pockets' chemical environment in class I HLA; correlated preferences (e.g. jointly hydrophobic P2/P Ω in HLA-A*02:01) are precisely interaction contrasts, hence the PSSM AUROC 0.9156 versus CNN 0.9044 and GNN 0.8480 in §5 is a dataset-specific comparison, not a theoretical ceiling or proof of interaction variance.

3 Appendix B: Hydrophobic moment properties

$\mu_H(x; \delta) = (1/N) |\sum_{n=1..N} H_{\{x_n\}} e^{i n \delta}|$, $\delta = 100^\circ$.

Proposition 3. (i) $0 \leq \mu_H \leq (1/N) \sum_n |H_{\{x_n\}}|$, equality iff all phasors align ($H_n e^{i n \delta}$ all same argument); (ii) μ_H is invariant under reversal combined with conjugation symmetry $|\mu_H(x)| = |\mu_H(\text{rev } x)|$; (iii) for a periodic amphipathic helix with hydrophobic moment phase at multiples of δ -period, μ_H is maximized by placing the largest $|H_n|$ at aligned phases.

Proof. (i) is the triangle inequality with equality condition. (ii) reversal maps the sum to $e^{i(N+1)\delta}$ times the conjugate sum, same modulus. (iii) rearrangement inequality applied to the aligned components. $\square$

4 Appendix C: PPV under class imbalance

With binder prevalence π and a score s with conditional binder density $f_1(s)$, non-binder $f_0(s)$, the posterior is $p(s) = \pi f_1(s) / (\pi f_1(s) + (1-\pi) f_0(s))$. Top- k PPV estimates $E[p \mid s \text{ in top-}k]$. For the benchmark's choice $k = \#\{\text{true binders}\}$, $\text{PPV}(k) \rightarrow \sup$ over threshold policies of precision at recall $= \pi \cdot N / |\{\text{predicted}\}| \dots$ (operational point used by MHCflurry's published benchmark, which we follow for comparability).

4.1 Locked G3 allele-held-out challenge: negative result

The committed `results/g3_allele_holdout.json` records 83,333 training examples and 6,000 evaluations across 12 test alleles. Overall AUROC was **0.6687** for ours versus **0.9285** for MHCflurry. Mean per-allele AUROC was **0.5798** versus **0.9056**. The recorded paired two-sided Wilcoxon p -value across the twelve allele AUROCs is 0.00048828125 in favor of the comparator; gate G3

is false. This differs from the peptide-held-out results above. HLA-C*06:02 has only 40 rows (ours 0.3324, comparator 0.8242), so its individual comparison is unstable, but it does not overturn the large aggregate gap. No allele-generalization or leader-beating claim follows from the earlier within-dataset ensemble AUROC of 0.9273. [PENDING: independent allele-held-out improvement and prospective assay validation.]

4.2 Paper-build correction log, September 27, 2026

An earlier draft claimed that marginal residue coverage and large n guaranteed additive-model identifiability, and incorrectly asserted that the observed design covariance was block diagonal. Both claims are now withdrawn. Appendix A states the exact observed-design rank condition without claiming an empirical audit. The Appendix's earlier CNN AUROC of 0.9057 conflicted with its results table (0.9044) and is corrected. The G3 negative is preserved rather than recast as success.

5 Evidence-locked companion analysis: binding and CPP screening

This companion expands the paper with committed outputs only. It does not rerun the science builder or turn retrospective comparisons into new experiments. Each section names its JSON source. The earlier within-allele and peptide-held-out results must be read alongside the independently failed G3 allele-held-out test.

5.1 Matched-row binding benchmark: what actually happened

The committed `results/head_to_head_mhcflurry.json` evaluates 6,000 examples, 5,057 called binders under its defined threshold. High binder prevalence makes positive predictive value less revealing than AUROC or partial AUROC. The comparison uses predictions on the same recorded rows, not proof that each model had identical training access.

Model	AUROC	AUROC at FPR ≤ 0.1	PPV
MHCflurry	0.9164	0.5762	0.9383
Per-allele PSSM	0.9202	0.5676	0.9425
CNN	0.9026	0.5127	0.9343
PSSM/CNN ensemble	0.9281	0.6026	0.9440

On these selected same rows, ensemble AUROC exceeds MHCflurry by 0.0118 in the stored point summaries. The committed `results/h2h_bootstrap.json` reports 2,000 resamples, a mean delta 0.01171 and a percentile 95% interval [0.00430, 0.01918]. This conditions on the sample and the evaluation protocol. It is not an independent held-out-allele replication, as G3 demonstrates. The bootstrap reports $p_{\text{win}}=1$ across the recorded resamples, which is not a literal probability of superiority in all populations.

The `results/g2_wilcoxon.json` per-allele comparison has 30 evaluable alleles and 21 ensemble wins; mean AUROC delta 0.0130, median 0.0110 and one-sided Wilcoxon $p=0.0104$. Alleles are not independent biological cohorts, so this test does not license a clinical-generalization claim. The sign and size of a small within-panel margin should be judged against the much larger held-out-allele failure.

5.2 Locked G3 transfer failure, allele by allele

The separate `results/g3_allele_holdout.json` has 83,333 training examples, 6,000 evaluations and 12 held-out test alleles. These test alleles are not the same unit of generalization as a peptide-disjoint split within familiar alleles. G3 overall AUROC is 0.6687 versus 0.9285 for MHCflurry; mean per-allele AUROC 0.5798 versus 0.9056. Its recorded two-sided paired Wilcoxon p is 0.00048828125, favoring the comparator, and the gate is false.

Held-out allele	n	Our AUROC	MHCflurry AUROC
HLA-A*02:01	2290	0.6794	0.9304
HLA-A*02:03	933	0.6534	0.9142

Held-out allele	n	Our AUROC	MHCflurry AUROC
HLA-A*26:01	512	0.5553	0.9365
HLA-A*32:01	154	0.6634	0.8301
HLA-A*33:01	510	0.6557	0.8967
HLA-A*69:01	365	0.6030	0.9496
HLA-B*08:03	28	0.4470	0.8864
HLA-B*15:03	113	0.5498	0.8879
HLA-B*27:05	359	0.6007	0.9337
HLA-B*39:01	234	0.5238	0.9330
HLA-B*58:01	462	0.6937	0.9444
HLA-C*06:02	40	0.3324	0.8242

All twelve recorded per-allele comparisons favor MHCflurry numerically. The 28-row HLA-B*08:03 and 40-row HLA-C*06:02 strata are noisy on their own, but the pooled and mean-per-allele gaps remain large. No data-dependent exclusion of those small alleles is justified after seeing the result. [PENDING: an architecture and training protocol locked before a new unseen-allele benchmark.]

5.3 Assay-method slices and confounding limits

The committed `results/assay_method_headtohead.json` reports three assay-method strata. Their listed row counts do not sum to all 6,000 rows, so these are a subset view rather than an exhaustive decomposition; the file also records exact replication of its subset calculation (maximum difference zero).

Method	n	Binders	MHCflurry	Ensemble	Delta
Competitive radioactivity	4056	3677	0.8856	0.9002	0.0147
Direct fluorescence	1166	823	0.9539	0.9599	0.0060
Competitive fluorescence	757	548	0.9208	0.9230	0.0022

The three listed differences are positive but small. Multiple assay systems differ in measurement scale and chemistry. Any claim about a mutation mechanism requires assay-aware evaluation, not a post-hoc positive slice. `results/binder_confounds.json` fits a model to the 16,870-row held-out peptide panel controlling for length, assay method and locus; the ensemble coefficient is 1.3411 per standardized score with recorded odds ratio 3.823 and standard error 0.0214. This is association conditional on the recorded covariates, not causal proof or transfer validation. Its McFadden pseudo- R^2 changes from 0.0852 in the covariate-only model to 0.5008 when the score enters. Outcome-derived evaluation and feature construction must remain separated in any follow-up.

5.4 Calibration and decision thresholds

The committed `results/calibration_decomposition.json` records $n=16,870$, base rate 0.2998 and binned Brier score 0.09839; a constant-base-rate predictor has 0.20991. Reliability term

0.000597 and resolution 0.11211 are binned decomposition quantities, not a guarantee of calibration under a new allele or clinic. Expected calibration error in `results/calibration.json` is 0.0164 for its bins. A threshold selected after observing a held-out cohort would invalidate a prospective PPV claim. [PENDING: predeclared decision threshold and calibration on new assay/allele domains.]

5.5 Computational CPP nominations and safety boundary

The committed `results/cpp_novel_candidates.json` records 50 screened sequences, 18 meeting its implemented novelty filter, and ten sequence families. This is a filter result, not proof of biological novelty, delivery or safety. The highest-score sequences are often rich in Arg/Trp/Lys; without measured uptake and toxicity, classifier score alone is not an actionable treatment ranking.

ID	Score	Charge	Moment	Max Jaccard
9B-CPP-1	0.9979	10.0	0.635	0.176
9B-CPP-2	0.9965	11.0	0.811	0.429
9B-CPP-3	0.9964	8.0	0.420	0.333
9B-CPP-4	0.9963	7.0	1.197	0.222
9B-CPP-5	0.9962	9.0	0.482	0.250
9B-CPP-6	0.9961	7.0	0.669	0.444
9B-CPP-7	0.9957	8.1	0.241	0.267
9B-CPP-8	0.9955	5.0	0.836	0.286
9B-CPP-9	0.9955	11.0	0.859	0.286
9B-CPP-10	0.9955	7.1	0.250	0.286
9B-CPP-11	0.9954	11.0	0.266	0.273
9B-CPP-12	0.9954	9.0	0.493	0.214
9B-CPP-13	0.9953	7.0	0.399	0.308
9B-CPP-14	0.9953	10.1	1.158	0.273
9B-CPP-15	0.9953	12.0	0.947	0.316
9B-CPP-16	0.9952	10.1	1.064	0.400
9B-CPP-17	0.9952	11.0	0.332	0.300
9B-CPP-18	0.9951	11.2	1.511	0.222

Sequences, in the same order as the table:

- 9B-CPP-1: LRILRWWRWRWRNRWKRRR
- 9B-CPP-2: RLRRRLRLRLRLRLRLRL
- 9B-CPP-3: WRWRKKKKRKK
- 9B-CPP-4: GLWRWRWRWRKSLKK
- 9B-CPP-5: RRINRRWRWKRWKKC
- 9B-CPP-6: WRWRKKRRR

- 9B-CPP-7: RGRWRWKKRHRPK
- 9B-CPP-8: GLARWRWRWRQ
- 9B-CPP-9: RLRRRRRRNRNRNWRWL
- 9B-CPP-10: WRWKWKRWKRKH
- 9B-CPP-11: RLFRRRRLRRRRRRNN
- 9B-CPP-12: RLRLRLRLRLRIRIR
- 9B-CPP-13: AWRWRWRCKAKKR
- 9B-CPP-14: RRRWRRLWRHLRRR
- 9B-CPP-15: YLLRLRRRLRRRSRARRRR
- 9B-CPP-16: RGRLRRHLRRRLRRR
- 9B-CPP-17: RLRRRLRLRRFLRRRR
- 9B-CPP-18: LGRLWRRLRRRRRHLRRAARSH

Several candidates share basic-rich motifs and may be near-duplicates under a biological similarity metric even when a small reference search passes. The published `max_db_identity` entries for these candidates can be high; a pass against a defined reference database is not a global novelty claim. [PENDING: measured cellular uptake with endosomal-escape and membrane-damage controls, hemolysis testing, a broad homology audit and an assay-aware independent test.]

5.6 Evidence map and limits

- `results/phla_benchmark.json`: 114,688 measured (peptide, allele) pairs and 60 alleles in the draft protocol. Its peptide-disjoint split supports within-panel generalization, not unseen-allele performance.
- `results/head_to_head_mhcflurry.json` and `results/h2h_bootstrap.json`: same-row comparison and paired empirical uncertainty under the selected test panel.
- `results/g2_wilcoxon.json`: per-allele within-panel comparisons. This is descriptive against the independent transfer failure.
- `results/g3_allele_holdout.json`: locked negative transfer test; it overrides any blanket leader-beating assertion.
- `results/cpp_benchmark.json` and `results/cpp_novel_candidates.json`: classifier and screen outputs, not wet-lab evidence.

These artifacts support a scoped scientific contribution: a transparent baseline-plus-ensemble panel, a failure mode exposed by allele-held-out evaluation, and testable computational CPP nominations. They do not establish a best-in-world immune prediction tool, peptide therapy, or clinical performance.